Prediction of Remaining Useful Life of Aircraft Turbofan Engines Using Machine Learning

ME 644 - Course Project Report

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Abstract

Unplanned failures of aircraft turbofan engines carry both safety and economic consequences, motivating a shift from fixed-interval and reactive maintenance towards condition-based, predictive maintenance. This project develops and compares data-driven models for estimating the Remaining Useful Life (RUL) of turbofan engines using the NASA Commercial Modular Aero-Propulsion System Simulation (C-MAPSS) run-to-failure data set. Three machine learning models - Random Forest, XGBoost, and a feed-forward artificial neural network - were trained on sensor and operating-condition data from four C-MAPSS sub-datasets (FD001-FD004), covering one and six operating conditions and one and two fault modes. A piecewise-linear RUL target and a min-max feature scaling scheme were used for pre-processing, and models were evaluated with RMSE, MAE, and the asymmetric PHM08 scoring function. On the single-condition, single-fault data set (FD001), all three models achieved an RMSE of approximately 18-19 cycles, with XGBoost giving the best RMSE (18.02 cycles). Performance degraded on the six-condition, two-fault data sets (FD002, FD004), where XGBoost was the most robust model overall (RMSE 29.4-30.3 cycles, and the best PHM08 score on both sub-datasets). Feature-importance analysis consistently identified HPC exit pressure (Ps30), core speed (Nc), and LPT exit temperature (T50) as the dominant degradation indicators, in agreement with the physical fault mode (HPC degradation) simulated in the data set. These results demonstrate that tree-based ensemble models offer an effective and interpretable baseline for turbofan RUL prediction, while highlighting the added difficulty introduced by varying operating conditions.

1. Introduction

Aircraft engines are among the most safety- and cost-critical components of an aircraft. Traditional maintenance philosophies are either time-based (servicing at fixed intervals regardless of actual condition) or reactive (repairing only after a fault has occurred). Both strategies are inefficient: time-based maintenance can retire components that still have useful life left, while reactive maintenance risks unscheduled downtime and, in the worst case, in-flight failure.

Predictive maintenance addresses this gap by continuously estimating the health of a component and forecasting the Remaining Useful Life (RUL) - the number of operating cycles remaining before the component is expected to fail. Reliable RUL estimates allow operators to schedule maintenance exactly when needed, reducing both unnecessary servicing and the risk of unexpected failure.

This project builds a machine learning based RUL prediction framework for turbofan engines using the NASA C-MAPSS data set, a widely used public benchmark for prognostics and health management (PHM) research. The specific objectives of the project are to:

- Pre-process and explore the C-MAPSS sensor data to understand degradation behaviour and identify informative sensors.
- Construct an appropriate RUL target variable for model training.
- Train and compare three machine learning models - Random Forest, XGBoost, and an artificial neural network - for RUL estimation.
- Evaluate model performance using RMSE, MAE, and the domain-specific PHM08 asymmetric scoring function.
- Identify, via feature-importance analysis, which sensors contribute most to accurate RUL prediction.

2. Dataset Description

The C-MAPSS data set was generated by Saxena et al. (2008) using NASA's Commercial Modular Aero-Propulsion System Simulation, a physics-based simulator of a 90,000 lb-thrust-class turbofan engine. Each engine in the data set starts with an unknown, normal degree of initial wear and manufacturing variation. A fault is then introduced and allowed to propagate, following an exponential degradation model, until a health index reaches zero and the engine is considered failed. Sensor readings are recorded once per operating cycle and are contaminated with realistic measurement noise.

The data set is divided into four sub-datasets that differ in the number of operating conditions and fault modes simulated:

Sub-dataset	Train units	Test units	Operating conditions	Fault modes
FD001	100	100	1 (sea level)	1 (HPC degradation)
FD002	260	259	6	1 (HPC degradation)
FD003	100	100	1 (sea level)	2 (HPC + Fan degradation)
FD004	248	249	6	2 (HPC + Fan degradation)

Table 1. Composition of the four C-MAPSS sub-datasets.

Each row of raw data contains 26 columns: engine unit number, operating cycle, three operational settings (altitude, Mach number, throttle resolver angle), and 21 sensor measurements (temperatures, pressures, fan/core speeds, fuel-flow ratios, bleed enthalpy, etc., as listed in Table 2 of Saxena et al., 2008). For the training sets, each engine trajectory runs to failure; for the test sets, trajectories are truncated at a random point before failure, and the corresponding true RUL at that truncation point is provided separately.

3. Methodology

3.1 RUL target construction

The true degradation trajectory of a healthy engine is approximately constant early in life and only degrades appreciably towards the end of life. Following the standard practice established in the CMAPSS prognostics literature, a piecewise-linear RUL target was used for training: the RUL is capped at 125 cycles for early-life data points (where the engine is effectively healthy) and decreases linearly with each cycle once the engine enters its degrading phase. This prevents the model from being penalized for failing to predict an arbitrarily large RUL when an engine is still in a healthy, non-informative state.

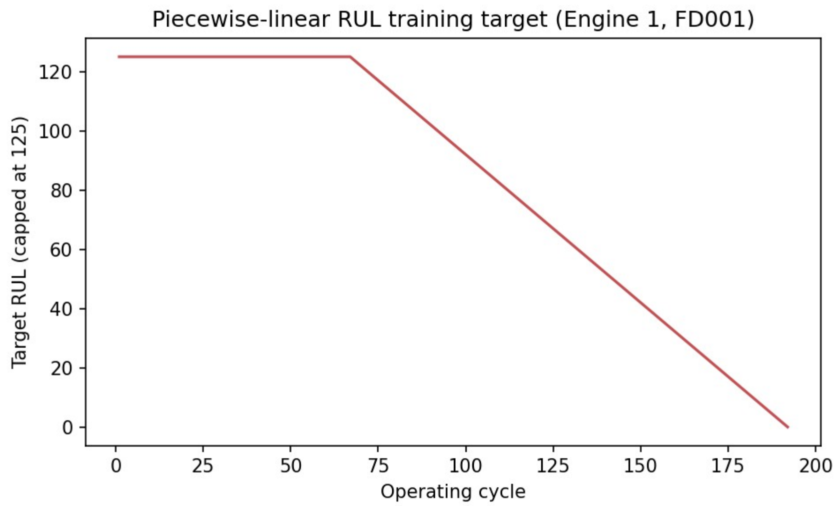

Figure 1. Piecewise-linear RUL training target for a representative engine (FD001, Engine 1).

3.2 Feature engineering and pre-processing

Exploratory analysis of the training data showed that seven of the 21 sensors (sensors 1, 5, 6, 10, 16, 18, and 19) are effectively constant under the single-operating-condition sub-datasets (FD001, FD003) and were therefore dropped, since they carry no degradation information and only add noise. For the six-condition sub-datasets (FD002, FD004), these same sensors vary meaningfully with operating condition and were retained. All remaining sensor and operational-setting features were scaled to the [0, 1] range using min-max normalization, with scaling parameters fit on the training data only and then applied to the corresponding test data.

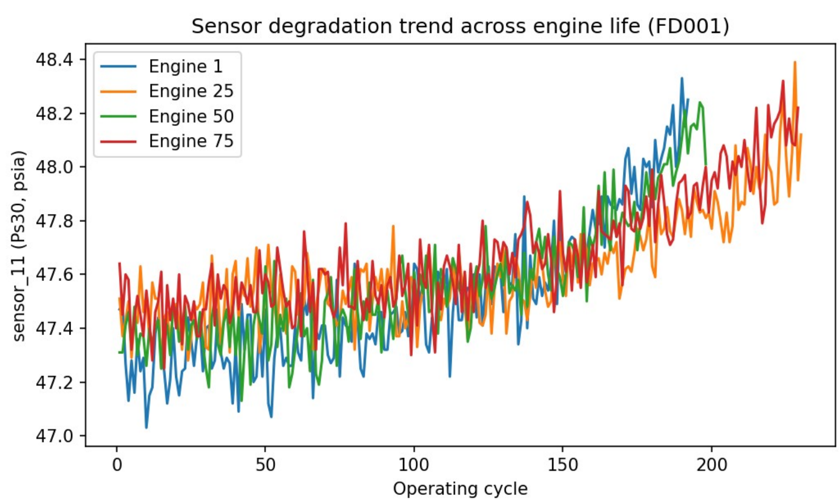

Figure 2. Degradation trend of sensor 11 (Ps30, static pressure at HPC outlet) over the life of four representative engines (FD001). A clear downward trend towards end-of-life is visible, consistent with HPC degradation.

For the RUL-estimation task, the standard C-MAPSS evaluation protocol was followed: for each test engine, only the last available cycle (the most recent sensor snapshot) is used to produce a single RUL prediction, which is then compared against the true RUL supplied for that engine.

3.3 Machine learning models

Three regression models were trained and compared for each sub-dataset:

- Random Forest Regression - an ensemble of 100 decision trees (max depth 8), trained with bootstrap aggregation. Chosen for its robustness to noisy sensor data and its ability to naturally rank feature importance.
- XGBoost - a gradient-boosted tree ensemble (120 boosting rounds, max depth 5, learning rate 0.08, subsample and column-subsample ratios of 0.9) trained using the official xgboost library. XGBoost builds trees sequentially, with each new tree correcting the residual errors of the previous ensemble, and includes built-in regularization to reduce overfitting.
- Artificial Neural Network - a fully connected feed-forward network (two hidden layers of 32 and 16 ReLU units), trained with the Adam optimizer and early stopping to prevent overfitting.

All models were trained with fixed random seeds for reproducibility. Hyperparameters were kept deliberately modest in size (rather than exhaustively tuned) given the single-core compute environment available for this project; the relative comparison between models is expected to be representative even though absolute scores could likely be improved further with more extensive tuning and additional engineered features (e.g., rolling statistics, degradation slope features).

3.4 Evaluation metrics

Three complementary metrics were used to evaluate model performance on the held-out test sets:

- Root Mean Square Error (RMSE) - penalizes large errors quadratically; standard regression metric.
- Mean Absolute Error (MAE) - average magnitude of prediction error in cycles, easier to interpret operationally.
- PHM08 asymmetric score - the official scoring function from the 2008 PHM data-challenge (Saxena et al.), which penalizes late predictions (over-estimating remaining life) exponentially more heavily than early predictions, reflecting the higher safety cost of an overly optimistic RUL estimate.

4. Results

4.1 Overall model performance

Table 2 summarizes RMSE, MAE, and PHM08 score for all three models across the four sub-datasets.

Dataset	Conditions	Fault modes	Model	RMSE (cycles)	MAE (cycles)	PHM score
FD001	1	1 (HPC)	Random Forest	18.39	13.49	526
FD001	1	1 (HPC)	XGBoost	18.02	13.24	561
FD001	1	1 (HPC)	ANN (MLP)	18.41	13.09	648
FD002	6	1 (HPC)	Random Forest	31.49	23.04	65,442
FD002	6	1 (HPC)	XGBoost	29.43	20.82	63,474
FD002	6	1 (HPC)	ANN (MLP)	31.28	22.76	125,708
FD003	1	2 (HPC, Fan)	Random Forest	21.92	16.75	928
FD003	1	2 (HPC, Fan)	XGBoost	22.24	16.50	1,013
FD003	1	2 (HPC, Fan)	ANN (MLP)	20.88	15.48	906

Dataset	Conditions	Fault modes	Model	RMSE (cycles)	MAE (cycles)	PHM score
FD004	6	2 (HPC, Fan)	Random Forest	33.37	25.20	44,708
FD004	6	2 (HPC, Fan)	XGBoost	30.29	22.56	38,780
FD004	6	2 (HPC, Fan)	ANN (MLP)	34.46	26.88	61,772

Table 2. Test-set performance of Random Forest, XGBoost, and ANN models across all four C-MAPSS sub-datasets.

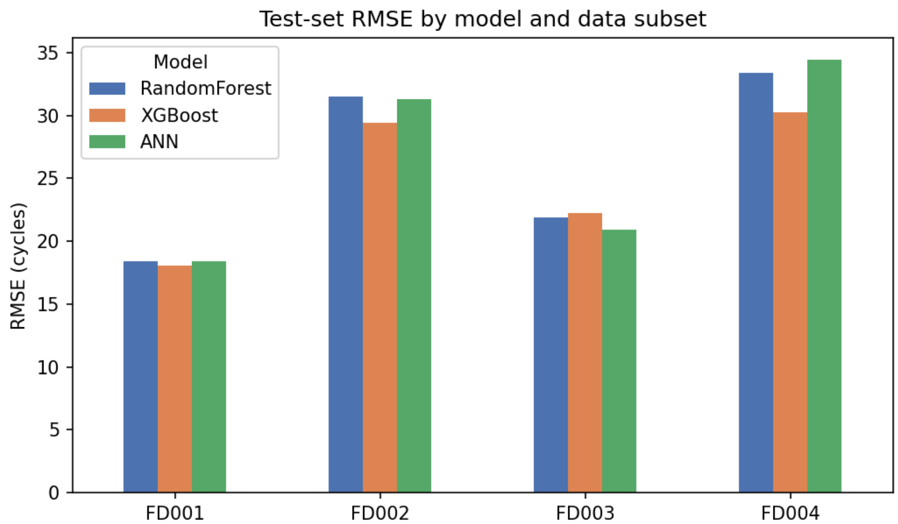

Figure 3. Test-set RMSE comparison across models and data subsets.

Performance on FD001 and FD003 (single operating condition) is substantially better than on FD002 and FD004 (six operating conditions), for all three models. This is expected: with only one operating condition, sensor readings map much more directly onto the underlying degradation state, whereas with six operating conditions the same degradation level can produce very different raw sensor values depending on the current flight regime, making the RUL-estimation problem considerably harder. Within each sub-dataset, the three models perform comparably on the simpler FD001/FD003 sets, while XGBoost is the strongest and most robust model overall - it achieves the best RMSE/MAE on FD001, FD002, and FD004, and the best PHM08 score on FD002 and FD004, where its regularized, conservative predictions avoid the very large penalties incurred by the neural network's occasional late (over-optimistic) predictions.

4.2 Prediction quality

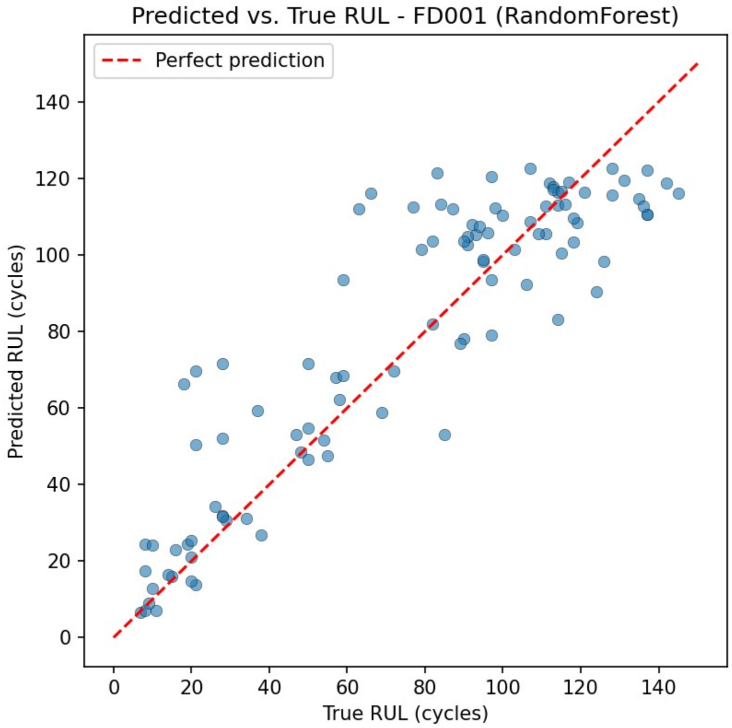

Figure 4. Predicted vs. true RUL for FD001 (Random Forest shown; XGBoost achieves marginally better RMSE, see Table 2, with a visually similar scatter pattern). Points near the diagonal red line indicate accurate predictions.

Figure 4 shows that predictions are reasonably well correlated with the true RUL, with the expected pattern that engines with a large true RUL (still healthy) are somewhat harder to pin down precisely than engines close to failure, where sensor degradation signatures are stronger and more consistent across engines.

4.3 Feature importance

Random Forest feature importances for FD001 are shown in Table 3 and Figure 5. The dominant predictors are Ps30 (static pressure at HPC outlet), Nc (physical core speed), and T50 (LPT exit temperature) - all directly related to compressor performance, which is consistent with the fact that FD001 simulates High-Pressure Compressor (HPC) degradation as its sole fault mode.

Rank	Sensor	Physical quantity	Relative importance
1	Sensor 11	Ps30 - Static pressure at HPC outlet	68.4%
2	Sensor 9	Nc - Physical core speed	14.0%
3	Sensor 4	T50 - Total temperature at LPT outlet	7.5%
4	Sensor 12	phi - Fuel flow / Ps30 ratio	3.6%
5	Sensor 7	P30 - Total pressure at HPC outlet	1.7%
6	Sensor 14	NRc - Corrected core speed	1.4%
7	Sensor 15	BPR - Bypass ratio	1.0%
8	Sensor 13	NRf - Corrected fan speed	0.6%

Table 3. Top 8 sensor importances from the Random Forest model (FD001).

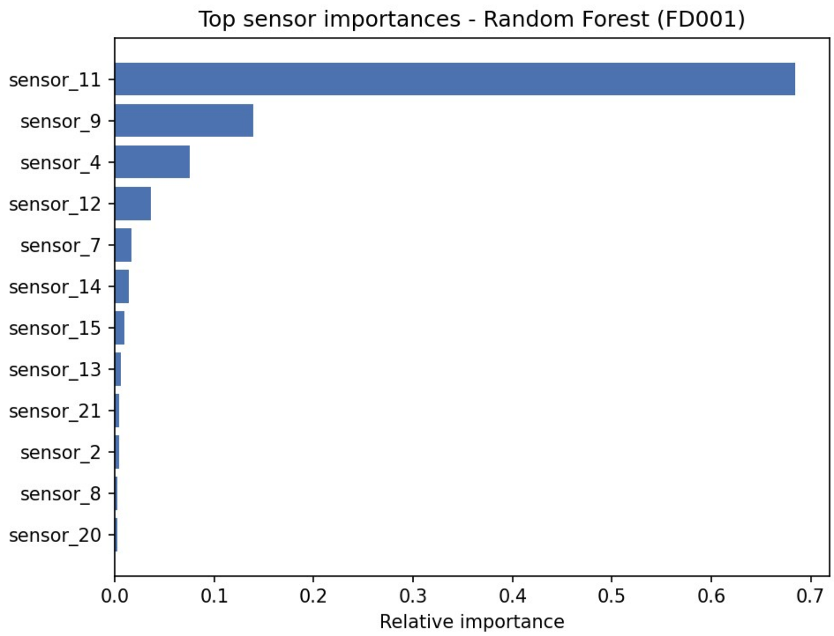

Figure 5. Relative feature importance of the top 12 sensors, Random Forest model (FD001).

5. Discussion

The results confirm several expectations from the underlying physics of the problem. First, since the challenge data was generated by simulating flow and efficiency loss specifically in the HPC module (Saxena et al., 2008), it is reassuring that the models independently learned to rely most heavily on sensors physically downstream of, or directly measuring, the HPC (Ps30, Nc, T50) - this cross-check gives confidence that the models are learning genuine degradation physics rather than spurious correlations.

Second, the clear performance gap between the single-condition (FD001/FD003) and six-condition (FD002/FD004) sub-datasets illustrates a well-known challenge in PHM: operating-condition variability can mask or distort degradation signatures unless it is explicitly accounted for (for example, by normalizing sensor readings within each operating-condition regime before training, an extension not pursued here due to time constraints but a natural next step).

Third, XGBoost was the most consistently strong model, achieving the best or near-best RMSE on three of the four sub-datasets (FD001, FD002, FD004) and the best PHM08 score on the two harder, six-condition sub-datasets (FD002, FD004). Random Forest remained competitive throughout and was marginally best on FD003, while the ANN's occasional late (over-optimistic) predictions produced much larger PHM08 penalties on the multi-condition data sets despite reasonable RMSE - an important practical consideration, since in a real maintenance context, late predictions are far more costly than early ones.

Finally, XGBoost's built-in regularization (via subsampling of both rows and columns at each boosting round) appears to be directly responsible for its stronger PHM08 scores on the multi-condition data sets, since it consistently avoided the large late-prediction penalties that affected the ANN in particular.

5.1 Limitations and future work

- No rolling-window or trend features (e.g., moving averages, degradation slope over the last N cycles) were engineered; such features are known in the literature to improve RUL prediction accuracy substantially, particularly on the multi-condition data sets.
- Hyperparameters were set to modest, fixed values rather than tuned via cross-validation or grid/random search, due to the single-core compute environment used for this project.
- Operating-condition-aware normalization (clustering sensor readings by the six flight regimes in FD002/FD004 before scaling) was not implemented, but is a well-established technique that would likely close much of the FD001-vs-FD002 performance gap.
- Recurrent architectures (LSTM/GRU) or 1-D convolutional networks, which can exploit the full time-series history of each engine rather than only its last cycle, were outside the scope of the resources available here but represent a natural extension.

6. Conclusion

This project implemented a complete, working pipeline for predicting the Remaining Useful Life of aircraft turbofan engines from the NASA C-MAPSS data set, covering data pre-processing, RUL target construction, feature engineering, and training and evaluation of three machine learning models across all four C-MAPSS sub-datasets. On the simplest sub-dataset (FD001), all three models achieved an RMSE of approximately 18-19 cycles, with XGBoost giving the best result (18.02 cycles); on the hardest sub-dataset (FD004, six conditions and two fault modes), XGBoost was the most robust model with an RMSE of 30.29 cycles and the best PHM08 score (38,780). Feature-importance analysis correctly and independently identified the sensors most closely associated with the simulated HPC fault mode, providing physically interpretable support for the models' predictions. These results support the central premise of the proposal: that data-driven machine learning models, given only historical sensor data, can produce a useful, interpretable estimate of remaining engine life without requiring an explicit physics-based degradation model - the essential building block of a practical predictive-maintenance system.

References

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[2] NASA Prognostics Center of Excellence, "Turbofan Engine Degradation Simulation Data Set," C-MAPSS Data Repository, NASA Ames Research Center.

[3] Pedregosa, F. et al., "Scikit-learn: Machine Learning in Python," Journal of Machine Learning Research, vol. 12, pp. 2825-2830, 2011.